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The Robotic Age

The world around us is constantly evolving in unprecedented ways, at an unimaginable speed. The robotic age, something we're used to reading about in science fiction is finally becoming a reality. The result? Hopefully, a robot in every house. Robots have a long history, from fictional characters (Isaac Asimov's novels and motion pictures), to industrial robots and mobile robots.

Industrial robots have taken a long stride and are well established, but newer domains and research has got most of the limelight. Roboticists are now taking the robots out of their fixed base (industrial applications) and are imparting mobility and intelligence. There are a number of mobile robots deployed in the real world.

Robotics is an exciting, multi-disciplinary area that integrates intelligent control, communications, computer vision, mechatronics, sensor fusion, design and many other aspects on a single platform. Although the vast majority of robots today are used in factories, advances in technology enable robots to automate many tasks in non-manufacturing industries such as agriculture, construction, health care and other services.

India is now emerging as a destination for the production of industrial robots. In fact, the size of the Indian robotics market is large with sales figures hovering around Rs. 3,500 crores. Most of this (around 70 per cent), is in the automotive industry. The robotics industry in India is expected to grow substantially in the years to come. Several new players

are venturing into this space. The Indian manufacturing arena is poised to grow larger in the coming decades, thereby increasing the demand for robotic solutions. As far as the industry is concerned, India contributes significantly as far as research / intellectual property (IP) generation to converting LISP codes to C codes for industrial robots is con-

cerned. A large volume of work centred around these domains is outsourced to India. Being an agrarian society, our nation has opportunities in automating agricultural related activities.

Industrial robotics is mature as a technology, and is well established. However, the mobile robotics space is expected to witness significant growth in the coming



ROBOTICS, INDIA AND THE FUTURE